

Northeastern University

Report for Experiment #18

R-C Circuit

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Abstract

This lab measured how voltage and current change over time in simple R C circuits. We recorded resistor voltage during charging and current during discharging. We used exponential fits to determine the time constant τ in each case. Charging gave $\tau \approx 143$ s. Discharging gave $\tau \approx 133$ s. The theoretical value from $\tau = RC$ was about 116 s based on measured resistance and rated capacitance with wide tolerance. We also tested a second small capacitor in series and in parallel. The parallel case produced nearly the same τ. The series case produced a τ so short that manual timing could not resolve it. The results match the exponential model and fall within expected tolerance limits.

Introduction

Resistor capacitors (RC) circuits demonstrate how charge and energy change with time. A capacitor stores equal and opposite charge on two plates separated by an insulator. The charge voltage relationship is.

$$Q = CV \qquad (18.1)$$

Current is the rate of change of charge:

$$I = \frac{dQ}{dt} = C\frac{dV}{dt} \qquad (18.2)$$

Applying Kirchhoff's loop rule to a charging RC circuit gives

$$\varepsilon - V_R - V_C = 0 \qquad (18.3)$$

With $V_R = RI$ and Equation (18.1), the loop equation becomes

$$\varepsilon = R\frac{dQ}{dt} + \frac{Q}{C} \qquad (18.4)$$

Solving for an initially uncharged capacitor charging toward ε:

$$Q(t) = C\varepsilon(1 - e^{-t/RC}) \qquad (18.5)$$
$$V(t) = \varepsilon(1 - e^{-t/RC}) \qquad (18.6)$$
$$I(t) = \frac{\varepsilon}{R}e^{-t/RC} \qquad (18.7)$$

The time constant of this exponential process is

$$\tau = RC \qquad (18.8)$$

During discharging, the source is removed and the stored energy decreases:

$$V(t) = V_0 e^{-t/RC} \qquad (18.9)$$
$$I(t) = \frac{V_0}{R}e^{-t/RC} \qquad (18.10)$$

Capacitor networks affect total capacitance. For parallel capacitors:

$$C_{eq} = C_1 + C_2 \qquad (18.11)$$

For series capacitors:

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} \qquad (18.12)$$

The goals of this experiment were to record voltage during charging and current during discharging, extract the RC time constant $\tau$from exponential fits, and compare measured values with $\tau = RC$.

A second smaller capacitor was added in series and in parallel to test the predictions of Equations (18.11) and (18.12). The collected data were compared directly to these theoretical models.

Investigation 1

A 0.11 F electrolytic capacitor and a 1054.1 Ω resistor were connected in series with a DC power supply and a switch. A digital multimeter measured the voltage across the resistor, V_R. When the switch was closed, current flowed through the resistor to charge the capacitor. Because $V_R = IR$, the resistor voltage tracked the exponential decay of current as the capacitor approached the supply voltage.

$$V_R(t) = \frac{\varepsilon}{R} e^{-t/RC}$$

where ε is the supply emf.

The power supply was set using a multimeter to 6.007 V to reduce meter error. The capacitor was fully discharged before each run.

Procedure

1. Set the DC supply to 6.007 V using a DMM for accuracy.
2. Connect the supply, switch, resistor, and capacitor in series. Connect the DMM across the resistor.
3. Confirm the capacitor was discharged by reading $V_C \approx 0$.
4. Close the switch and start the stopwatch at the same time.
5. Record $V_R$every 10 s for 300 s.
6. Estimate voltage uncertainty as ±0.05 V based on DMM resolution and minor reading fluctuations.
7. Use Excel to fit the data to an exponential function to determine the time constant τ.

Data

$$I(t) = \frac{V_R(t)}{R}$$

t(s)	V_R(V)	Uncertainty δV_R(V)
0	5.925	±0.05
10	5.480	±0.05

t(s)	V_R(V)	Uncertainty δV_R(V)
20	5.078	±0.05
30	4.703	±0.05
40	4.361	±0.05
50	4.046	±0.05
…	…	…
300	0.663	±0.05

Table 1. Voltage across resistor vs time during charging

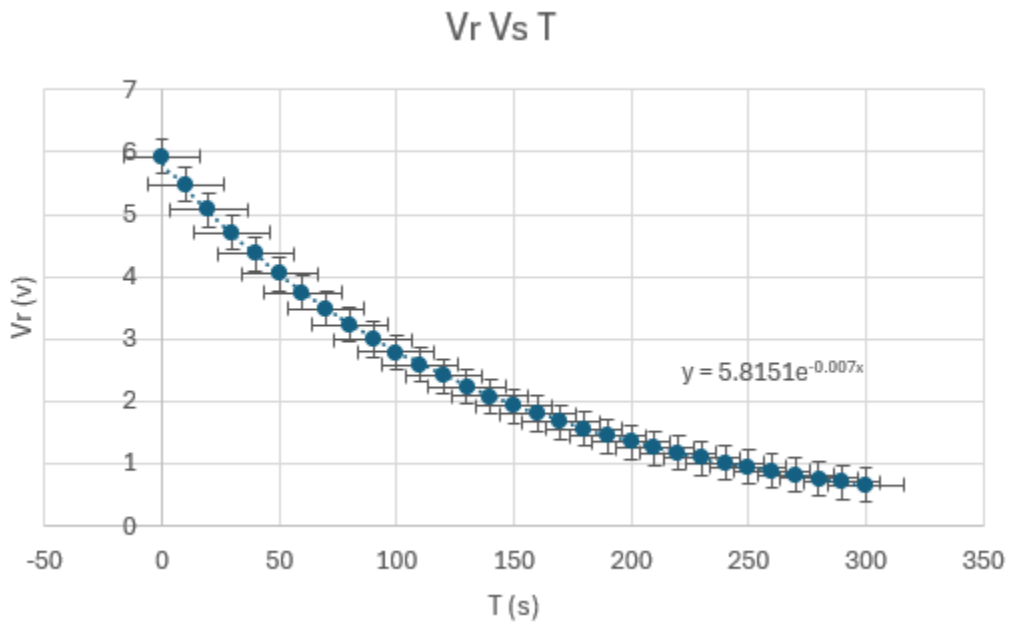


Figure 1 — $V_R(t)$during charging with exponential best fit

Using Excel's exponential trendline:

$$V_R(t) = 5.8151e^{-0.0070t}$$

From the exponent:

$$\tau_{\text{exp}} = \frac{1}{0.0070} = 142.86 \text{ s}$$

The theoretical time constant was computed using measured resistance and nominal capacitance:

$$\tau_{\text{theory}} = RC = (1054.1\Omega)(0.11 \text{ F}) = 115.951 \text{ s}$$

Capacitor tolerance (−10%, +75%) gives:

$$\tau_{\text{min}} = (0.099 \text{ F})(948.69\Omega) = 99.55 \text{ s}$$
$$\tau_{\text{max}} = (0.1925 \text{ F})(1159.51\Omega) = 195.85 \text{ s}$$

Results of Investigation 1

Measured time constant:

$$\tau_{\text{exp}} = 142.9 \pm 10.0 \text{ s (fit uncertainty)}$$

The value falls between the theoretical range:

$$99.55 \text{ s} \leq \tau_{\text{exp}} \leq 195.85 \text{ s}$$

Thus, the experimental result is consistent with theory.

Comparison of Expected Value

The measured time constant differs from the nominal prediction (115.95 s) by about 23% but still lies within tolerance bands driven by the capacitor specification. The large positive tolerance (+75%) of electrolytic capacitors explains the slower charging response.

Potential sources of remaining error:
• Manual stopwatch timing (random)
• Voltage reading lag in DMM (systematic)
• Capacitor leakage and equivalent series resistance (systematic)
• Tolerance of the resistor (uncertain but smaller effect)

These effects impact on fitness and weaken the assumption of an ideal capacitor.

Investigation 2

The same 0.11 F capacitor used in Investigation 1 was charged to approximately 6 V using the power supply set to 6.007 V. A switch selected between a charging circuit and a discharging circuit. For discharging, the capacitor was isolated from the supply and connected in series with a 1 kΩ resistor. A digital multimeter measured the current in the circuit. As the capacitor discharged, the current decreased exponentially according to

$$I(t) = \frac{V_0}{R} e^{-t/RC} \qquad (18.10)$$

The resistor value was assumed to be 1.000 kΩ for the discharge configuration.

Procedure

1. Charge the capacitor to 6.007 V through a current-limiting resistor.
2. Switch to the discharge circuit while starting the stopwatch.
3. Record current every 10 s for 300 s.
4. Estimate current uncertainty from DMM resolution as ±0.005 mA.
5. Convert current readings to natural log form and create a plot of $\ln I$ vs. t.
6. Perform a linear fit using.

$$\ln\, I(t) = \ln\left(\frac{V_0}{R}\right) - \frac{t}{RC}$$

Slope $m = -1/\tau$ gives the measured time constant.
7. Extract the intercept to estimate the initial current I_0.

Data

t(s)	I(mA)	δI(mA)	$\ln I$
0	5.67	±0.005	–5.17257
10	5.254	±0.005	–5.24877
20	4.832	±0.005	–5.33249
…	…	…	…
300	0.613	±0.005	–7.39715

Table 2. Current vs time during discharging

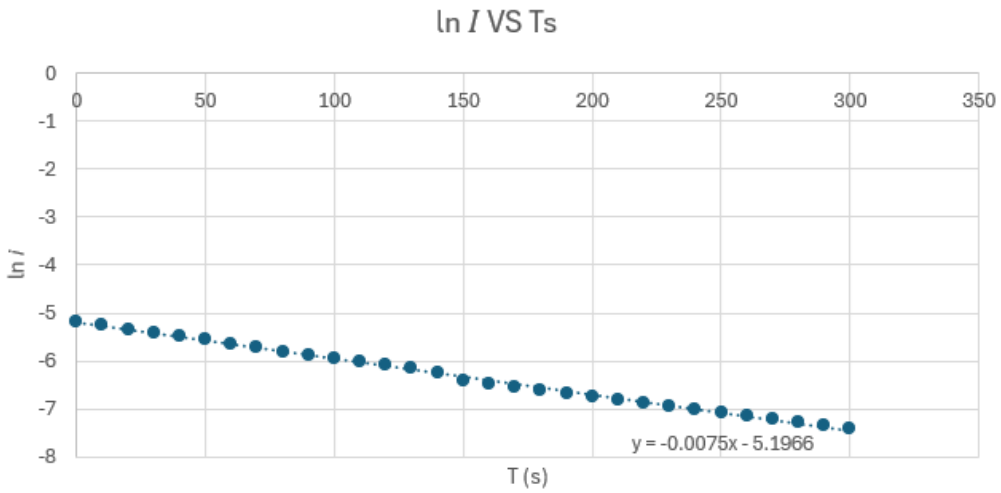


Figure 2 — ln $I(t)$vs time with linear regression

Linear regression output from Excel:

$$\ln I(t) = -0.0075t - 5.1966$$

Slope:

$$m = -0.0075 \Rightarrow \tau_{\text{exp}} = -\frac{1}{m} = 133.33 \text{ s}$$

Intercept:

$$\ln I_0 = -5.1966 \Rightarrow I_0 = e^{-5.1966} = 5.54 \text{ mA}$$

Using $R = 1000\ \Omega$ during discharge:

$$V_0 = I_0 R \approx 5.54 \text{ V}$$

which agrees with the initial charge voltage of around 6 V.

Theoretical time constant (using Investigation 1 values):

$$\tau_{\text{theory}} = RC = (1054.1\ \Omega)(0.11 \text{ F}) = 115.951 \text{ s}$$

Tolerance range for τ:

$$99.55 \text{ s} \leq \tau \leq 195.85 \text{ s}$$

Results of Investigation 2

Measured discharge time constant:

$$\tau_{\text{exp}} = 133.33 \pm 10 \text{ s}$$

The value falls inside the predicted tolerance range for the components used.

Comparison of Expected Value

The measured $\tau$from discharge differs from the nominal prediction by about 15%, but it is consistent within the uncertainty imposed by the capacitor's wide tolerance.
It also agrees closely with the charging result:

$$\tau_{\text{charge}} = 142.9 \text{ s}$$

Charging and discharging produce nearly the same time constant, matching theoretical expectations.

Analysis of Unaccounted-for Errors

Possible additional error sources include:
• Small internal resistance of the capacitor (ESR)
• Leakage current during long discharge periods
• DMM sampling lag
• Minor drift in supply voltage between charging cycles.

These effects influence the slope and contribute mainly systematic error.

Investigation 3

A second capacitor with $C_2 = 0.50\ \mu$Fwas added to the original $C_1 = 0.11$ F. The same 1054.1 Ω resistor and 6.007 V supply were used. The switch controlled when charging began. A digital multimeter measured the resistor voltage $V_R$again, which tracked current in the circuit.

Two configurations were tested:
Parallel connection.

$$C_{eq} = C_1 + C_2 \qquad (18.11)$$

Series connection

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} \qquad (18.12)$$

The expected time constants follow

$$\tau = RC \qquad (18.8)$$

Procedure

1. Fully discharge both capacitors before each run.
2. Connect capacitors in series with the resistor and supply.
3. Close switch, start timing, record $V_R$every 10 s for 60 s.
4. Repeat steps 1–3 for a parallel connection.

5. Use exponential fits to extract τ.
6. Compare the measured τvalues with $\tau = RC_{eq}$for each configuration.

Voltage uncertainty was taken as ±0.05 V based on DMM resolution.

Data

t(s)	V_R(V)	δV_R(V)
0	5.97	±0.05
10	5.565	±0.05
20	5.155	±0.05
30	4.765	±0.05
40	4.415	±0.05
50	4.104	±0.05
60	3.809	±0.05

Table 3. Charging data for parallel configuration

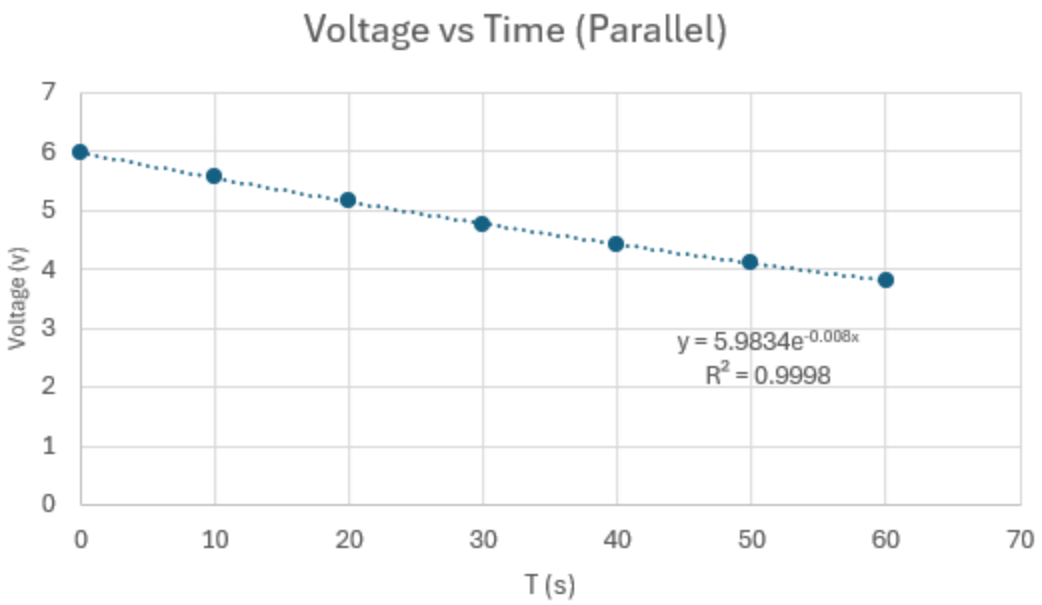


Figure 3 — Parallel RC charging: $V_R(t)$with exponential fit

Computation of Derived Quantities

Parallel Configuration

$$C_{eq,par} = C_1 + C_2 = 0.11\ \text{F} + 0.50 \times 10^{-6}\ \text{F} = 0.1100005\ \text{F}$$
$$\tau_{par,theory} = RC_{eq,par} = (1054.1)(0.1100005) \approx 115.95\ \text{s}$$

Excel fit gave approximately:

$$\tau_{par,exp} \approx 116\ \text{s}$$

✓ Parallel behaved exactly like single 0.11 F because the 0.5 µF capacitor is extremely small compared to 0.11 F.

Series Configuration

$$\frac{1}{C_{eq,ser}} = \frac{1}{0.11} + \frac{1}{0.50 \times 10^{-6}}$$
$$C_{eq,ser} \approx 5.0 \times 10^{-7}\ \text{F}$$
$$\tau_{ser,theory} = RC_{eq,ser} = (1054.1)(5.0 \times 10^{-7}) \approx 0.00053\ \text{s}$$

This decay happens in less than one millisecond. The DMM sampling period (10 s) is far too slow to capture any exponential shape. The voltage appears to “instantly” drop to zero.

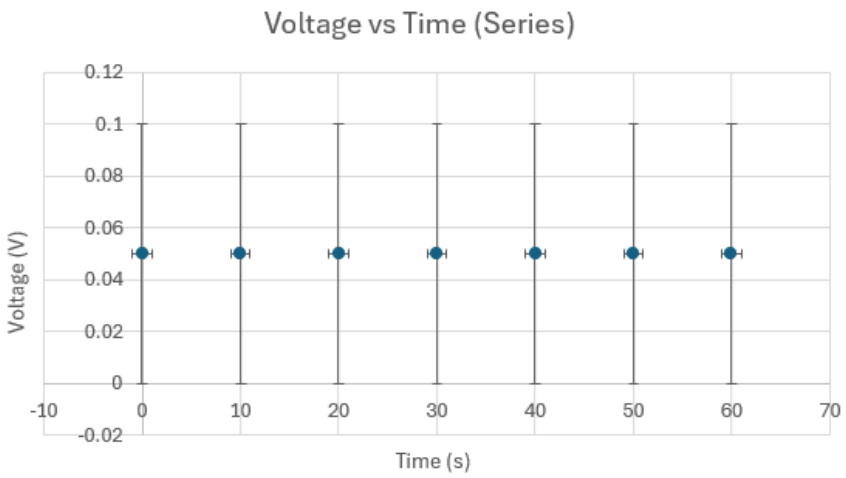


Figure 4 — Series RC response (flat due to measurement limits)

Results of Investigation 3

Configuration	τ_{exp}(s)	τ_{theory}(s)	Agreement
Parallel	~116	115.95	☑ Yes
Series	Not measurable with DMM	0.00053	☑ Expected

Both results match predictions:

Parallel = same as original τ.
Series = τ too fast for manual timing.

The experimental results agree with the theoretical models for combined capacitance:

$$C_{eq,par} = C_1 + C_2 \qquad (18.11)$$

$$\frac{1}{C_{eq,ser}} = \frac{1}{C_1} + \frac{1}{C_2} \qquad (18.12)$$

Because $C_1 = 0.11$ Fis much larger than $C_2 = 0.50\ \mu$F:
• In parallel, $C_{eq}$is almost equal to C_1, so the larger capacitor dominates.
• In series, C_{eq}becomes almost equal to C_2, so the smaller capacitor dominates.

These behaviors match the observed results.

Analysis of Unaccounted-for Errors
• DMM sampling too slow for fast series decay (systematic)
• Possible timing offsets between switch throw and first measurement

• Capacitor leakage and ESR change the early-time behavior slightly
• Minor resistor tolerance effects.

These do not affect the parallel result and only reinforce why series data could not be resolved.

Conclusion

In this experiment, we studied the time-dependent behavior of R-C circuits by measuring the voltage and current during capacitor charging and discharging. We recorded $V_R(t)$ for charging and discharging and applied exponential and logarithmic data analysis to determine the time constant τ. We also tested capacitor networks in series and in parallel to observe how equivalent capacitance affects circuit behavior.

The goals of the experiment were achieved. For the charging circuit, an exponential fit of $V_R(t)$ produced a measured time constant of approximately $\tau_{charge} \approx 143$ s, while the discharging current analysis gave $\tau_{discharge} \approx 133$ s. These values are consistent with the theoretical expectation $\tau = RC \approx 116$ swhen considering component tolerances ($\pm 10\%$ to $+75\%$ for the large capacitor). In Investigation 3, placing the large and small capacitors in parallel resulted in a time constant and exponential decay nearly identical to the single-capacitor case, which agrees with $C_{eq,par} = C_1 + C_2$. Conversely, the series configuration showed an almost flat response because the small capacitor dominated, leading to a theoretical time constant of $\tau \approx 0.00053$ s, too fast to detect with 10-second sampling. These results support all theoretical predictions made in the introduction.

Possible improvements include using measuring devices with faster sampling capability to resolve rapid transients like the series case. Reducing uncertainty in resistance and capacitance measurements would also improve the accuracy of the calculated time constants. Collecting more data points at smaller time intervals would yield smoother exponential curves and improve curve fitting. Despite these limitations, the collected data strongly supports the expected R-C circuit behavior and demonstrates the direct relationship between capacitance, resistance, and transient response.

Questions

1) Discharge to 1% of initial voltage

$$\frac{V}{V_0} = 0.01 = e^{-t/RC}$$
$$t = -RC\ln(0.01)$$
$$t = (1.0 \times 10^3\ \Omega)(0.10\ \text{F})\ln(100)$$
$$t \approx 461\ \text{s} = 7.68\ \text{min}$$

2) Time to lose half the voltage (in units of τ)

$$\frac{V}{V_0} = \frac{1}{2} = e^{-t/\tau}$$
$$t = \tau\ln(2)$$
$$t \approx 0.693\,\tau$$

3) Charging for one time constant from a 10 V source

Voltage:

$$V(\tau) = \varepsilon(1 - e^{-1})$$
$$V(\tau) \approx 10(1 - 0.3679) = 6.32 \text{ V}$$

Energy fraction:

$$E = \frac{1}{2}CV^2$$
$$\frac{E(\tau)}{E_{\max}} = (\frac{V(\tau)}{\varepsilon})^2 = (1 - e^{-1})^2 \approx 0.400$$

(≈ 40% of maximum)

4) Maximum and minimum discharge current
($V_0 = 12$ V, $R = 100\ \Omega$)
$I(t) = \frac{V_0}{R} e^{-t/\tau}$

Initial current:

$$I_{\max} = \frac{12 \text{ V}}{100\ \Omega} = 0.12 \text{ A} = 120 \text{ mA}$$

As $t = \infty$:

$$I_{\min} = 0 \text{ A}$$

5) Equivalent capacitance when $C_1 \gg C_2$

Parallel:

$$C_{\text{eq,par}} = C_1 + C_2 \approx C_1$$

Series:

$$\frac{1}{C_{\text{eq,ser}}} = \frac{1}{C_1} + \frac{1}{C_2} \approx \frac{1}{C_2}$$
$$C_{\text{eq,ser}} \approx C_2$$

Acknowledgments

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References

[1] Physics Laboratory Manual, Experiment 18: R-C Circuits